



Analog CMOS Circuit Design SS26

Practical Exercise II – Amplifier Testbenches

Group

Name Zhanyu Dong Sumit Kumar Patel

Matriculation Number 6159975 5992625

Date 31/5/2026

Feedback

.....

.....

.....

.....

.....

.....

.....

Points **Mark**

Date **Signature**

No review without this standard form!



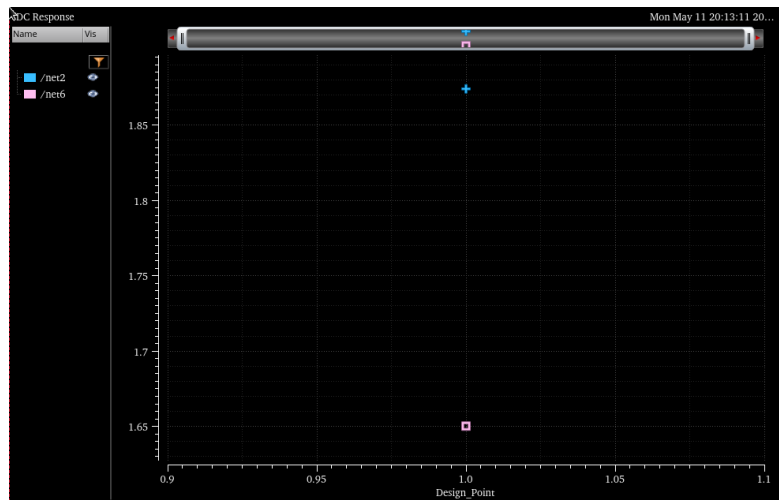
May 31, 2026

1 T1: DC Bias Point and Offset

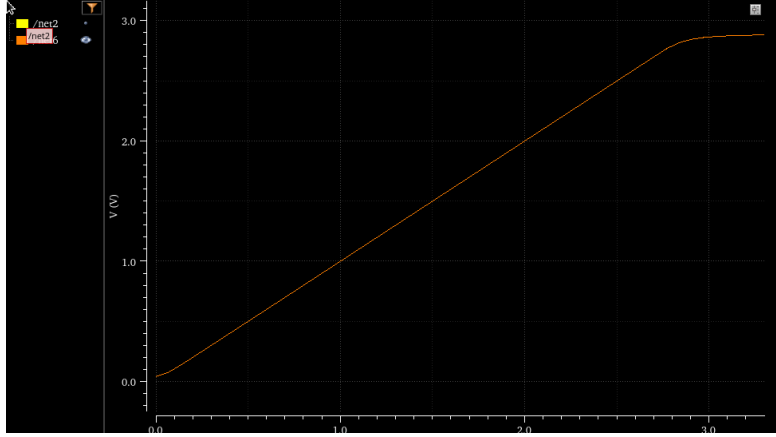
1.1 a

$$V_{\text{DC}}(a) = 1.874085 \text{ V}$$

$$V_{\text{DC}}(b) = 1.650211 \text{ V}$$



1.2 b



$$V_{\text{ICM,min}} = V_{\text{THn}} + V_{\text{ov1,2}} + V_{\text{ov7}}$$

$$V_{\text{ICM,max}} = V_{\text{DD}} - V_{\text{ov3,4}} - V_{\text{ov5,6}} - V_{\text{THp}}$$

1.3 c

$$(V_{\text{DC}}(\text{"/net6"}) - V_{\text{DC}}(\text{"/net3"})) V_{\text{offset}} = 210.7 \mu\text{V}$$

Yes, reasonable.

It arises from transistor mismatches, finite output resistance, and non-ideal current mirrors.

1.4 d

Implementation (b) uses *unity negative feedback*, which forces the op-amp to operate in its linear region. This prevents the output from saturating, allowing the DC offset to be measured directly as the difference between the output voltage and the input common-mode voltage. In contrast, the open-loop configuration (a) drives the output to the supply rail due to the high open-loop gain, making the offset impossible to quantify.

2 T2: AC Transfer Function

2.1 a

$$H(s) = \frac{v_{\text{out}}}{v_{\text{in}}} = \frac{A_{\text{DC}}}{1 + \frac{A_{\text{DC}} Z_1}{Z_1 + Z_2}}$$

2.2 b

$$Z_1 = \frac{1}{sC_1}$$

$$Z_2 = R_2$$

$$H(s) = \frac{A_{DC}}{1 + A_{DC} \cdot \frac{1/(sC_1)}{1/(sC_1) + R_2}} = \frac{A_{DC}}{1 + A_{DC} \cdot \frac{1}{1 + sC_1R_2}} = \frac{A_{DC}(1 + sC_1R_2)}{1 + sC_1R_2 + A_{DC}}$$

2.3 c

1 Zero

$$z = -\frac{1}{R_2C_1}$$

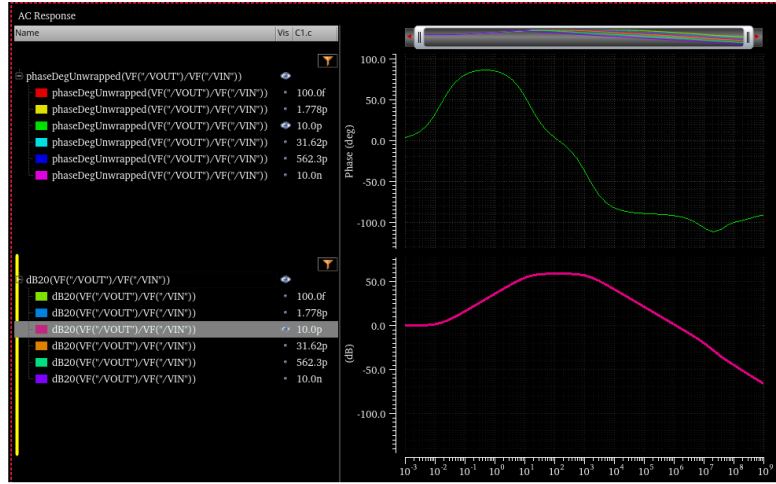
1 Pole

$$p = -\frac{1 + A_{DC}}{R_2C_1} \approx -\frac{A_{DC}}{R_2C_1}$$

2.4 d

$$R_2 = 100 \text{ M}\Omega, \quad C_1 = 100 \text{ nF}.$$

2.5 e



- **External zero:** $f_z \approx 0.159 \text{ Hz}$, introducing a +20 dB/dec slope in the magnitude response and contributing a 90 deg phase shift.
- **External pole:** $f_p \approx A_{DC} \cdot f_z \approx 14.1 \text{ Hz}$.
- **Internal amplifier pole:** Observed at $f_{p,\text{int}} \approx 13 \text{ kHz}$

2.6 f

- DC gain $y_{\max}(\text{dB20}((V_F("VOUT") / V_F("VIN"))))$
- UGBW $\text{unityGainFreq}((V_F("VOUT") / V_F("VIN")))$
- Phase Margin $\text{phaseMargin}((V_F("VOUT") / V_F("VIN")))$

2.7 g

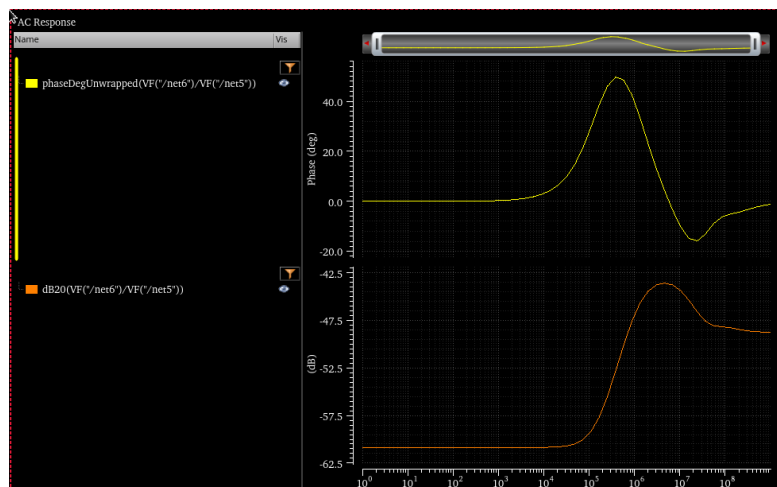
Since the amplifier has one single dominant pole, the load capacitance C_L can be arbitrarily large.

3 T3: Power Supply Rejection

3.1 a

- Parasitic capacitive coupling between the supply rails and the output node, bypassing the amplifier's feedback loop.
- Reduction of the loop gain at high frequencies, weakening the negative feedback that suppresses supply noise.
- Finite output impedance of current mirrors and bias circuits, which allows supply variations to modulate the tail current and appear at the output.

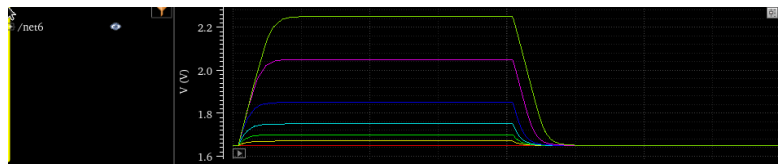
3.2 b



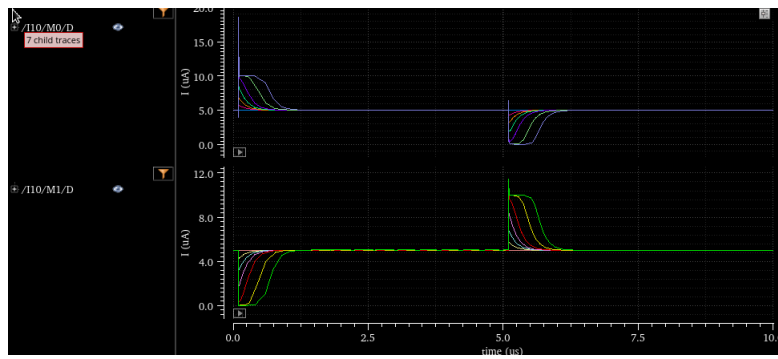
$$\text{PSR}_{\text{DC}} = 60.9 \text{ dB}$$

4 T4: Step Response

4.1 a

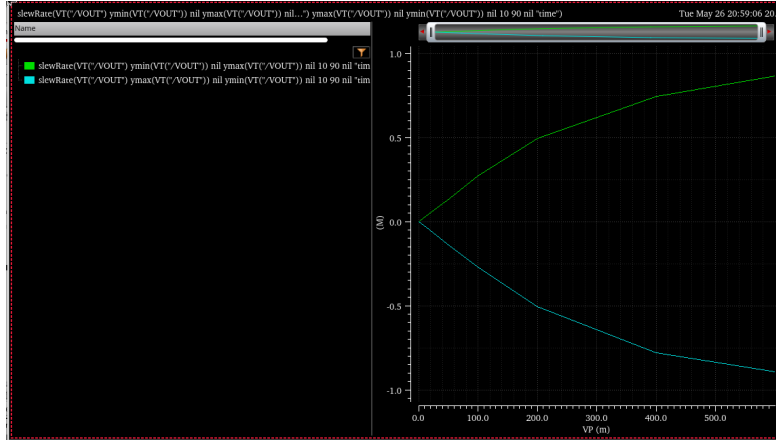


4.2 b



4.3 c

Rising edge: `slewRate(VT("/VOUT") ymin(VT("/VOUT")) nil ymax(VT("/VOUT")) nil 10 90 nil "time")`
 Falling edge: `slewRate(VT("/VOUT") ymax(VT("/VOUT")) nil ymin(VT("/VOUT")) nil 10 90 nil "time")`



4.4 d

Small-signal linear region: With small input steps, the input differential pair is not saturated. The slew rate increases proportionally with input amplitude, limited by small-signal bandwidth.

Slew-rate-limited large-signal region: With large input steps, the differential stage saturates. The internal maximum charge/discharge current for load capacitor is fixed, so the slew rate reaches a constant maximum value and no longer increases with larger input steps.

4.5 e

$$V_{Boundary} \approx 400mv$$